
Effects of nutrient solution formulas and concentrations on the growth rate and chlorophyll content of *Wolffia arrhiza*

Ruanto, P.¹, Chaichana, N.² and Tunma, S.^{3*}

¹Biology program, Faculty of Education, Chiang Rai Rajabhat University, 57100, Thailand;

²Science program, Faculty of Education, Chiang Rai Rajabhat University, 57100, Thailand;

³Chemistry program, Faculty of Education, Chiang Rai Rajabhat University, 57100, Thailand.

Ruanto, P., Chaichana, N. and Tunma, S. (2026). Effects of nutrient solution formulas and concentrations on the growth rate and chlorophyll content of *Wolffia arrhiza*. International Journal of Agricultural Technology 22(5):2407-2418.

Abstract This research investigated the optimal concentration of AB hydroponic fertilizer to maximize the growth and nutritional quality of watermeal (*Wolffia arrhiza*), a highly productive aquatic plant. Given its potential as a sustainable resource, identifying precise nutrient thresholds is critical for efficient large-scale cultivation. The results demonstrated that nutrient concentration significantly dictated biomass outcomes and biochemical composition. The highest productivity was observed at a concentration of 1000 ppm, which produced a maximum dry weight of 38.08 ± 4.67 g. This concentration also optimized the plant's photosynthetic capacity, yielding the highest average amounts of chlorophyll a (40.81 ± 1.91 mg/L) and chlorophyll b (8.40 ± 1.91 mg/L). Furthermore, statistical evaluation revealed a significant linear relationship between pigment levels and biomass. Chlorophyll content was identified as a reliable factor for predicting the total yield of *Wolffia arrhiza*. Through linear regression analysis, a predictive model was developed, allowing for the estimation of harvest yields based on these biochemical markers. This research provided a quantitative framework for optimizing hydroponic systems, ensuring high yields while managing resource inputs efficiently for sustainable agricultural applications.

Keywords: Watermeal, Hydroponic fertilizer, Chlorophyll, Growth

Introduction

Wolffia arrhiza, commonly called watermeal, is the smallest flowering plant species. It is green throughout, has no root, and is around 0.1 centimeters in length. It grows in natural still water sources and has been more widely consumed in the northern and northeastern regions of Thailand. Typically, it floats on or just below the water surface but only for short periods. The thallus is round or oval in shape, can be found either solitary or in pairs, thick on both the top and bottom or sometimes flat on top and thick on the bottom. New thallus can be produced within the original ones. The flowers are extremely small and

* **Corresponding Author:** Tunma, S.; **Email:** Somruthai.tun@crru.ac.th

clustered within a pouch at the edge of the thallus. Each cluster contains a single male flower, consisting of one stamen with a single anther, and a single female flower, consisting of one ovary with a single ovule (Daengsawat, 2001). The seeds are very small. *Wolffia arrhiza* is the only plant in the Lemnaceae family that can be consumed by humans as it contains up to 20% protein and 40% carbohydrates by dry weight and also has other beneficial nutrients (Appenroth *et al.*, 2017). Research has shown that it can reach full maturity in just 4 days (Sripen, 1999).

The majority of watermeal produced from natural water sources is not in a continuous flow and may be exposed to various chemical contaminants, including both insecticides and herbicides. Moreover, the contamination hinders its growth and reduces its yield. To solve this problem, watermeal is cultivated in clean water ponds to control water quality and nutrition, which are crucial factors affecting the growth of watermeal. Typically, organic or synthetic granular fertilizers containing major nutrients (N, P, K) and lime are used. Improper fertilizer application or poor fertilizer solubility can lead to fertilizer precipitation, rendering the nutrients unavailable for the plants, thus affecting their growth (Sricharoen *et al.*, 2001). Therefore, this research is aimed at applying different formulas of hydroponic fertilizers, which are water soluble and contain essential organic nutrients necessary for plant growth. Due to their liquid state, hydroponic fertilizers can readily dissolve in water, making the nutrients immediately available to the plants without the risk of fertilizer precipitation or microbial contamination, which is commonly associated with organic fertilizers. Therefore, different concentration ratios of hydroponic fertilizers are used for cultivating watermeal to compare the yield quantity and quality. This approach aimed to mitigate the issue of inconsistent watermeal production in nature and the contamination of harmful chemicals that may pose risks to consumers.

Materials and methods

Fertilizer preparation

Fertilizers for hydroponic cultivation of watermeal consist of Stock A and Stock B, prepared separately in the form of nutrient solutions. Stock A contains calcium oxide (CaO) at a concentration of 2.0%, and iron (Fe) at a concentration of 0.05%. They are essential for the stronger structure of plant cell walls, aiding in protein synthesis, increasing fruit size, being a component of various enzymes, and playing a role in the plant's respiration process. Stock B contains magnesium oxide (MgO) at a concentration of 1.0% and sulfur (S) at a concentration of 0.5%,

along with micronutrients including 0.05% iron (Fe), 0.20% manganese (Mn), 0.05% copper (Cu), 0.05% zinc (Zn), and 0.05% boron (B). These elements are components of chlorophyll used by plants in photosynthesis, which help plants appear green and grow properly. They also facilitate the movement of phosphorus within the plant, assist in protein synthesis, enhance the rate of photosynthesis, and are components of various enzymes that support the plant's respiration process.

Sample preparation

Plastic tanks were filled with clean water and left for one night. Then, the watermeal sample culture was added into the plastic tanks. Other plant contaminants, such as small duckweed, were removed from the tanks, leaving only the watermeal samples. After 5 minutes, only those floating on the water surface were collected and weighed to obtain a quantity of 100 grams for further use as samples to cultivate watermeal with different ratios of hydroponic fertilizer formulas.

Cultivation and data collection

Black plastic tanks, of 55-centimeter diameter, with a slant height of 30 centimeters, were used for cultivating the watermeal sample. Each tank was filled with 10 liters of clean tap water and left overnight to allow the chlorine to evaporate. Hydroponic fertilizer stock A and B at different concentrations were added. The experimental design used in this research was a triplicate experimental design conducted over a 14-day cultivation cycle. Each experimental condition was performed in three separate trials to ensure the reliability and consistency of the results. Four different treatments were listed as follows:

- Group 1 (Control): 0 ppm (No fertilizer)
- Group 2: 1000 ppm (1 ml of fertilizer A and B per 1 liter of water)
- Group 3: 2000 ppm (2 ml of fertilizer A and B per 1 liter of water)
- Group 4: 3000 ppm (3 ml of fertilizer A and B per 1 liter of water)

The pH of water was adjusted to between 5.5 and 6.0, using sulfuric acid and sodium hydroxide. Then, 25 grams of watermeal sample was added to each tank. The tanks were covered with fine mesh to prevent mosquitoes and insects from laying eggs. Water was replaced every 7 days by discarding the old water and refilling water containing the same ratio of fertilizer, before reintroducing the watermeal. The data was collected daily, including the light intensity, pH

values, temperature, electrical conductivity (EC), and the total dissolved solids (TDS). After 14 days, the total wet mass of watermeal in each experimental group data was measured. On the final day of cultivation, the watermeal from each experimental group was removed and weighed to determine the wet weight. Then, the aluminum container containing the sample was dried by a hot air oven at 105 degrees Celsius for approximately 3 hours. The container was removed from the oven to let it cool in a desiccator for about 30 minutes at room temperature, weighed, then dried for additional 30-minute intervals until a constant dried weight was achieved. The chlorophyll a and b content of a 100 mg sample was also analyzed by extracting with ethanol until the tissue color turned white. The solution was adjusted to a total of 30 ml volume for measuring the absorbance of the ethanol-extracted chlorophyll solution at wavelengths of 645 and 663 nm using a spectrophotometer. The obtained values were used to calculate the amounts of chlorophyll a and chlorophyll b, based on a modification of Arnon's method (1949). The absorbance was measured at wavelengths of 663 and 645 nm, and the chlorophyll content was calculated using the following formulas:

$$\text{Chlorophyll A (g/L)} = 0.0127(A_{663}) - 0.00269(A_{645})$$

$$\text{Chlorophyll B (g/L)} = 0.0229(A_{663}) - 0.00468(A_{645})$$

where A663 and A645 are the absorbance values at wavelengths of 663 and 645 nm, respectively.



Figure1. Cultivation of *Wolffia arrhiza* samples

Regression analysis

Multiple Regression Analysis (MRA) is used to determine the relationship between the growth of watermeal (Y = total yields) and independent

variables (X_1 = chlorophyll quantity, X_2 = Total dissolved solid, X_3 = Electric conductivity, X_4 = pH value). The regression equation is determined using the Data analysis Add-in of Excel. The standard error and the determination coefficient are also calculated to identify the highest possible linear relationship between the independent variables and the dependent variable (Hair *et al.*, 1988).

Results

The yield of Wolffia arrhiza cultivated with fertilizer A and B at different concentration ratios

The *Wolffia arrhiza* cultivation was conducted at the average light intensity of approximately 7,400 lux in four experimental groups to compare the effect of using fertilizer formulas A and B at different concentration ratios to water. The data of wet weight were collected every 7 days, as shown in Table 1.

Table 1. Average total yield from each experimental group

Experimental group	Total yield (g)			Trendline equation
	Day 0	Day 7	Day 14	
Group 1 (No fertilizer)	25 ± 0.0 ^a	34.23 ± 6.82 ^a	42.28 ± 2.70 ^a	$y = 15.469\ln(x) + 24.598$
Group 2 (1 ml A+B/1 L water)	25 ± 0.0 ^a	43.83 ± 2.23 ^b	50.14 ± 4.23 ^b	$y = 23.345\ln(x) + 25.714$
Group 3 (2 ml A+B/1 L water)	25 ± 0.0 ^a	35.58 ± 5.41 ^a	39.12 ± 4.33 ^{ac}	$y = 13.112\ln(x) + 25.402$
Group 4 (3 ml A+B/1 L water)	25 ± 0.0 ^a	31.21 ± 2.96 ^a	34.76 ± 2.56 ^c	$y = 8.892\ln(x) + 25.013$

Note Different letters in the vertical column indicate the significant difference at the 0.05 level

The highest average total yield on day 7 was in Group 2 (1 ml fertilizer/1 L water). On day 14, the group with the greatest increase in yield (100%) was also Group 2, followed by Group 1 (69.12%) and 3 (56.48%), while the lowest increase was Group 4 (39.04%). The relationship between duration and total yield value was logarithmic with r^2 -value very close to 1.00. Moreover, comparisons of the average total yields of wet and dry weight from each experimental group after a 14-day cultivation period are presented in Table 2. The highest average total yield of dry weight (38.08 ± 4.67 grams) was also from Group 2, while the yield values of Groups 3 and 4 were not significantly different from the control group.

Table 2. the wet and dry weight of total yields from each experimental group

Experimental group	total yields (grams)	
	wet weight	dry weight
Group 1 (No fertilizer)	42.28 ± 2.70 ^a	24.59 ± 3.30 ^a
Group 2 (1 ml A+B/1 L water)	50.14 ± 4.23 ^b	38.08 ± 4.67 ^b
Group 3 (2 ml A+B/1 L water)	39.12 ± 4.33 ^{ac}	24.87 ± 2.08 ^a
Group 4 (3 ml A+B/1 L water)	34.76 ± 3.56 ^c	24.20 ± 4.80 ^a

Note Different letters in the vertical column indicate the significant difference at the 0.05 level

Data on the physical and chemical environmental conditions during the cultivation were collected. Light intensity was measured daily, as well as conductivity (EC), pH levels, temperature, and the amount of total dissolved solids (TDS). The recorded details are provided in Tables 3.

Table 3. Data recorded during the cultivation of *Wolffia arrhiza*

Properties	Recorded data ($\bar{x} \pm SD$)
Electric Conductivity: EC ($\mu S/cm$)	
Group 1 (No fertilizer)	120.97 ± 2.74 ^a
Group 2 (1 ml A+B/1 L water)	482.67 ± 1.89 ^b
Group 3 (2 ml A+B/1 L water)	876.34 ± 13.00 ^c
Group 4 (3 ml A+B/1 L water)	1,292.67 ± 16.50 ^d
pH values	
Group 1 (No fertilizer)	5.73 ± 0.58 ^{ab}
Group 2 (1 ml A+B/1 L water)	6.09 ± 0.13 ^b
Group 3 (2 ml A+B/1 L water)	5.71 ± 0.18 ^a
Group 4 (3 ml A+B/1 L water)	5.38 ± 0.22 ^a
Total Dissolved Solids: TDS	
Group 1 (No fertilizer)	77.34 ± 1.75 ^a
Group 2 (1 ml A+B/1 L water)	316.00 ± 9.43 ^b
Group 3 (2 ml A+B/1 L water)	560.67 ± 21.21 ^c
Group 4 (3 ml A+B/1 L water)	827.00 ± 10.37 ^d

During the experimental period, the average light intensity was approximately 7,400 lux and the average temperature was 25.7 °C. All the Electrical Conductivity (EC), pH and TDS values from all experimental groups increased from the initial day.

The contents of chlorophyll a and b obtained from Wolffia arrhiza cultivation

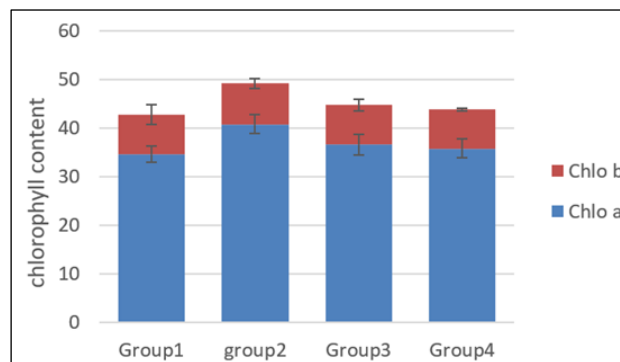
The average contents of chlorophyll a and b in the *Wolffia arrhiza* sample, cultivated with AB hydroponic fertilizer at different concentration ratios across all 4 groups, were measured over a total period of 14 days, as shown in Table 4. The highest average amount of chlorophyll a and b (40.81 ± 1.91 and 8.40 ± 1.91 mg/L, respectively) after 14 days of cultivation is from group 2. However, the amounts of chlorophyll b content from each group show no statistically significant difference at the 95% confidence level.

Table 4. The chlorophyll a and b contents of *Wolffia arrhiza* after 14 days

Experimental group	Chlorophyll content (mg/L)	
	Chlo a	Chlo b
Group 1 (No fertilizer)	34.66 ± 1.71 ^a	8.09 ± 2.02 ^a
Group 2 (1 ml A+B/1 L water)	40.81 ± 1.91 ^b	8.40 ± 0.95 ^a
Group 3 (2 ml A+B/1 L water)	36.59 ± 2.09 ^a	8.15 ± 1.15 ^a
Group 4 (3 ml A+B/1 L water)	35.78 ± 1.98 ^a	8.02 ± 0.35 ^a

Note Different letters in the vertical column indicate the significant difference at the 0.05 level

Total chlorophyll content and percentage comparisons of the average amounts of chlorophyll a and b across 4 groups of *Wolffia arrhiza* cultivated with hydroponic fertilizer formulas A and B at different concentration ratios is shown in Figure 2. Two types of chlorophyll were found in *Wolffia arrhiza* in a ratio of 4:1.

**Figure 2.** Chlorophyll a/b ratios across 4 groups of *Wolffia arrhiza* cultivated for 14 days

Regression analysis

The model for assessing the total yield of *Wolffia arrhiza* was developed from multiple regression analysis. The prediction equation for multiple regression analysis can be constructed as follows:

$$Y = \alpha + \beta_1 X_1 + \dots + \beta_k X_k$$

Regression models were developed to predict the quantities of total yield through Linear Regression Analysis, which is a method that evaluate the statistically significant relation between independent and dependent variables as shown in Table 5.

Where: Y represents the total yield of *Wolffia arrhiza* (dry weight).

X_1 represents total chlorophyll content.

X_2 represents Total Dissolved Solids.

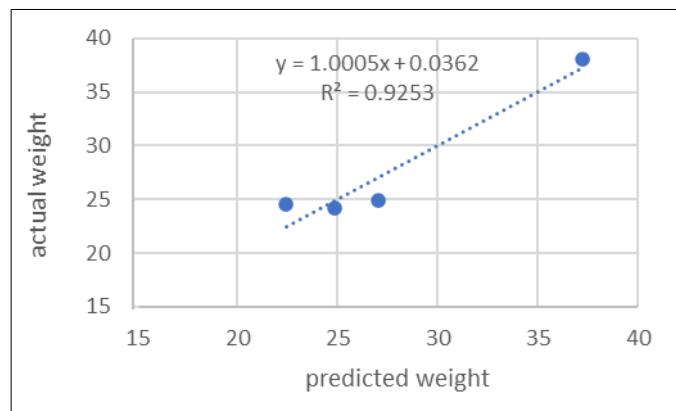
X_3 represents Electric Conductivity.

X_4 represents pH value.

Table 5. The models to predict total yield from different predictor variables

Predictor variables	R ²	Std. Error	Regression model for yield prediction	P-value
Total chlorophyll (X ₁)	0.93	2.26	$Y = -75.45 + 2.29X_1$	0.04
Total Dissolved Solids (X ₂)	0.08	7.94	- no relation -	
Electric Conductivity (X ₃)	0.09	7.92	- no relation -	
pH value (X ₄)	0.73	4.34	$Y = -85.96 + 19.89X_4$	0.15

The most suitable model, with a determination coefficient (R^2) of 0.93, was developed using chlorophyll content, serving as the only significant independent variable (p -value<0.05). The model was further tested for differences between actual and predicted values (Paired Sample T-Test). The results showed that the average actual and predicted values did not differ significantly at a 95% confidence level. Moreover, the relationship between actual dry weight and predicted dry weight can be obtained from the model as shown in Figure 3. The slope ($m=1.0005$) is extremely close to 1.0, indicating that when the predicted value from the model increases by 1 unit, the actual measured value also increases at an almost identical 1:1 ratio. The intercept ($c=0.0362$) is very close to 0, suggesting that if the model predicts a dry weight of 0, the actual biological dry weight measured would also be approximately 0.

**Figure 3.** The scatter plot of the actual versus the predicted dry weight of total yields obtained from the model

The R^2 value of 0.9253 indicates that 92.53% of the variation in actual dry weight can be explained by the predicted dry weight generated from this mathematical model (Figure 3). The remaining 7.47% may result from

experimental error or other factors not included in the model. This is consistent with the trendline, which closely follows nearly all observed data points.

Discussion

The research investigated the production of *Wolffia arrhiza* which cultivated with hydroponic fertilizers formula A and B at different concentration ratios in four groups (0 ml/1L (control group), 1 ml/1L, 2 ml/1L, and 3 ml/1L, respectively). Following a 14-day cultivation period, it can be concluded that the production of *Wolffia arrhiza* in group 2, cultivated with hydroponic fertilizers formula A and B at a concentration of 1 ml fertilizer/1L water, produced the highest wet and dry weights of 50.14 ± 4.23 and 38.08 ± 4.67 grams, respectively. In this experiment, data on the physical and chemical environmental conditions during the cultivation were collected. The average light intensity during cultivation was 7,400 lux. The water temperature was 25.65 degrees Celsius, on average. Group 1, 2, 3, and 4 had the average electric conductivity (EC) values of 142.83, 457.89, 836.44, and 1,211.78 $\mu\text{S/cm}$, respectively.

Group 2 had the highest yield of *Wolffia arrhiza* because of a growth environment that was chemically favorable for the species. The result showed that the average EC value of Group 2 was $482.67 \pm 1.89 \mu\text{S/cm}$, which is comparable to electrical conductivity measured in natural water sources that are favorable to *Wolffia arrhiza* growth (Ruekaewma *et al.*, 2015). A decreased production in Group 1 (control group) was due to its low average electrical conductivity ($120.97 \pm 2.74 \mu\text{S/cm}$). Groups 3 and 4 had average electrical conductivity values of 876.34 ± 13.00 , and $1,292.67 \pm 16.50 \mu\text{S/cm}$, respectively, which were considered too high, leading to the lowest yields. In order to exert pressure at the roots and facilitate the movement of nutrient-rich water from the roots to other sections, plants rely on transpiration through their leaves. If the EC value is too high, the plant won't be able to transport the nutrient-rich water to its various parts, resulting in poor growth and possible nutrient deficiencies (Ruekaewma, 2011). Additionally, the fertilizers are separated into two stocks since mixing certain chemicals from different stocks at high concentrations can cause precipitation. In hydroponic cultivation, which grows plants using water, the concentration and amount of the solution from both stock A and stock B can be controlled using an EC meter.

For hydroponic plant cultivation, the pH level is crucial for the chemical reactions with the nutrients used to nourish the plants. Group 1, 2, 3, and 4 had the average water pH values of 5.73 ± 0.58 , 6.09 ± 0.13 , 5.71 ± 0.18 , and 5.38 ± 0.22 , respectively. Naturally, acidic water aids in the good dissolution of nutrients, enabling plants to absorb and utilize them effectively (Piotrowska *et*

al., 2010). If the water used to combine plant nutrients is highly alkaline, it will precipitate and become unavailable for plant. Typically, the pH level for plant cultivation should range from 5.5 to 7.0, but the optimal pH for nutrient solubility is between 5.86 and 6.3 (Damna *et al.*, 2017).

The minerals and nutrients in the hydroponic fertilizer formula A and B were crucial chemical factors for the growth of *Wolffia arrhiza*. It is evident that the growth rate of *Wolffia arrhiza* in all four groups was impacted by the concentration of hydroponic fertilizer formula A and B. Group 2 produced the highest yield due to receiving an optimal amount of nutrients from the hydroponic fertilizer formula A and B. Group 1 (control group) had a lower yield because it did not receive nutrients from the fertilizer. Groups 3 and 4 produced the least yield due to receiving an excessive amount of nutrients, which slowed their growth. In addition, the ideal water temperature for freshwater plants to thrive is 25-30 degrees Celsius (Daengsawat, 2001). During the experiment, the average water temperature was 25.57 degrees Celsius.

In conclusion, the superior growth observed in Group 2 may be associated with the optimal electrical conductivity (EC) level of approximately 482 $\mu\text{S}/\text{cm}$, which likely promoted efficient nutrient uptake and cellular metabolism. Electrical conductivity is an important parameter in hydroponic cultivation because it reflects the concentration of dissolved ions available for plant absorption. Excessively low EC values may result in nutrient deficiency, while excessively high EC values can induce osmotic stress and reduce water absorption efficiency. In this study, Groups 3 and 4 exhibited substantially higher EC values, which may have inhibited nutrient transport and physiological activities, leading to lower biomass accumulation. In addition to EC, pH is another critical factor influencing nutrient solubility and bioavailability in hydroponic systems. The pH range observed in this experiment (5.38–6.09) was generally suitable for nutrient dissolution and uptake. However, Group 2 maintained pH values closer to the optimal range for hydroponic cultivation, which may have enhanced nutrient accessibility and metabolic efficiency. Previous studies have shown that slightly acidic conditions improve the solubility of essential nutrients, particularly micronutrients such as iron and magnesium, which are required for chlorophyll biosynthesis and photosynthetic activity.

Chlorophyll a and b levels in four experimental groups were compared following a 14-day cultivation period. The outcome demonstrated that the highest average amounts of chlorophyll a and b were found in Group 2 (40.81 ± 1.91 and 8.40 ± 0.95 mg/L, respectively), while Group 1 (control group) had the lowest levels, measuring 34.66 ± 1.71 and 8.09 ± 2.02 mg/L, respectively. The hydroponic fertilizer, particularly formula B, contains minerals and nutrients that are essential components of chlorophyll, including nitrogen, which plays a role

in plants' photosynthesis process, as well as magnesium and iron, which are crucial elements in chlorophyll synthesis. Plants that are deficient in these essential elements will not be able to produce enough chlorophyll, resulting in reduced photosynthesis and slower growth (Praphasert, 2007). Chlorophylls are key pigments that absorb light energy for use in the process of photosynthesis to promote plant growth (Aro and Andersson, 2003). Therefore, chlorophyll content can be utilized as a significant factor to predict the growth of *Wolffia arrhiza*, using the model developed through Linear Regression analysis. Although the predictive model developed in this study was based on a limited dataset, the findings suggest that chlorophyll measurement could potentially be used as a rapid preliminary indicator for estimating the productivity of *Wolffia arrhiza* under controlled cultivation conditions.

Chlorophyll accumulation was strongly associated with biomass production in *Wolffia arrhiza*. The highest chlorophyll a and chlorophyll b contents were observed in Group 2, corresponding to the highest dry weight yield. This relationship suggests that chlorophyll content may serve as an important physiological indicator of growth performance in watermeal cultivation systems. Chlorophyll molecules play a central role in light harvesting and photosynthetic energy conversion; therefore, increased chlorophyll concentration generally reflects enhanced photosynthetic capacity and biomass productivity. The presence of magnesium and iron in the nutrient solution may have contributed to increased chlorophyll synthesis because both elements are essential components involved in chloroplast formation and photosynthetic electron transport.

The present findings highlight the importance of optimizing nutrient management for the sustainable cultivation of *Wolffia arrhiza* as a future food resource. Watermeal has attracted increasing attention because of its rapid growth rate, high protein content, and potential application as an alternative sustainable crop for human consumption and animal feed production. Controlled hydroponic cultivation systems may provide a practical approach for improving biomass productivity while minimizing contamination risks associated with natural water sources. Therefore, nutrient optimization strategies could contribute significantly to the commercial-scale production of watermeal in the future.

Acknowledgements

This research project was supported by the faculty of education, and Research and Development Institute, Chiang Rai Rajabhat University.

Conflicts of interest

The authors declare no conflict of interest.

References

- Appenroth, K. J., Sree, K. S., Bohm, V., Hammann, S., Vetter, W., Leiterer, M. and Jahreis, G. (2017). Nutritional value of duckweeds (Lemnaceae) as human food. *Food Chemistry*, 217:266-273.
- Arnon, D. I. (1949). Copper enzymes in isolated chloroplasts. Polyphenoloxidase in *B. vulgaris*. *Plant Physiology*, 24:1-15.
- Aro, E. and Andersson, B. (2003). Regulation of photosynthesis. *Biologia Plantarum*, 46:382-382.
- Daengsawat, C. (2001). *Plant Physiology*. (2nd Edition). Department of Biology, Phetchabun Rajabhat University.
- Damna N., Saikaew W. and Tira-umphon A. (2017). Effect of fertilizer type and light filter level to yield and quality of wolffia [*Wolffia arrhiza* (L.) Wimm.]. *Songklanakarin Journal of Plant Science*, 4:60-64.
- Hair, J., Anderson, R., Tatham, R. and Black, W. (1988). *Multivariate Data Analysis*, 5th ed. Prentice-Hall. Inc. Upper Saddle River, NJ, USA.
- Piotrowska, A., Bajguz, A., Czerpak, R. and Kot, K. (2010). Changes in the growth, chemical composition, and antioxidant activity in the aquatic plant *Wolffia arrhiza* (L.) Wimm. (Lemnaceae) exposed to Jasmonic acid. *Plant Growth regulation*, 29:53-62.
- Praphasert, P. (2007). *Plant Physiology*. Bangkok: Odeon Store.
- Ruekaewma, N. (2011). Optimal conditions for production of Khai-nam *Wolffia globosa*. (Doctor's Thesis). Chulalongkorn University.
- Ruekaewma, N., Piyatiratitivorakul, S. and Powtongsook, S. (2015). Culture system for *Wolffia globosa* L. (Lemnaceae) for hygiene human food, *Songklanakarin Journal Science*, 37:575-580.
- Sricharoen, S., Charoenthetprasith, N., Jewyaem, W., Kaewborisut, S., Champasri, T. and Khongsai, S. (2001). Cultivation of Water Meal (*Wolffia arrhiza*) for Reducing Fish Feed Costs. *KKU Research Journal*, 6:6-15.
- Sripen, S. (1999). *Aquatic Plants in Thailand*. (1st Edition). Bangkok: Amarin Printing and Publishing.

(Received: 6 April 2025, Revised: 16 July 2026, Accepted: 13 September 2026)